ATHARVA PATIL

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Software Engineer · Founder · Product Builder

SUMMARY

Software engineer and product builder focused on desktop applications, developer tools, and systems programming. Founder of P4inz Interactive Labs, where I independently design, build, test, and ship privacy-first software: currently an encrypted vault, two Linux distributions, and a compiled programming language.

EXPERIENCE

P4inz Interactive Labs — Founder & Independent Software Studio

2026 – Present · Navi Mumbai (remote-friendly)

- Own the full product cycle myself on every project below: direction, architecture, engineering, UI/UX, testing, and release.
- Every product ships on the same principles: no telemetry, trade-offs explained on-screen, free wherever the studio can afford it to be.

Freelance — Web, UX & Brand

2025 – Present

- Independent client work in web development, UI/UX, and 3D/VFX visualization, including a full production website build for a performing-arts academy (kalasadhana-navimumbai.in).

SELECTED PROJECTS

Kanvaz — Visual Reference Workspace (Flagship, Free)

Jun 2026 – Present · Electron, MIT

- Built and maintain a free, MIT-licensed reference workspace for VFX and 3D artists: live 3D model preview (glTF, OBJ, FBX, STL, USD), a typed connection graph with a node-editor Map View, a Scratch Board drawing layer, and board templates, shipped for Windows, macOS, and Linux.
- Designed a local-only MCP Bridge that lets an AI agent read and edit the active board through a scoped tool set, off by default and fully undo-reversible.
- Shipped seven releases in about a week during one update cycle in September 2026 (v9.0.0 to v9.6.0), all backward-compatible with the free, open-source core.

Nexus — Encrypted Personal Vault

Mar 2026 · Windows Desktop · release on hold

- Engineered a Windows encrypted vault using AES-256-GCM authenticated encryption and Argon2id key derivation, with Windows Hello as a secondary unlock.
- Zero telemetry, zero cloud: content stays encrypted at rest and exists decrypted only in memory while the vault is unlocked.
- Rebuilt the entire v11 interface (lock screen, media browser, credentials, notes, settings, updater) without touching the cryptographic core underneath it.

MINK — Programming Language

2026 – Released v1.0.4 · Rust, Apache 2.0

- Designed and built a compiler from scratch: lexer, parser, type system, optimizer, and native code generator, producing standalone executables with no external toolchain.
- Added a second native backend (Linux ELF, alongside Windows PE), real threads, an async runtime, and a package manager, growing the regression suite to just under 3,000 native-execution tests.

Pursue OS — Investigation-Focused Linux OS

2026 – Present · Linux, Apache 2.0

- Built a Linux OS for OSINT, DFIR, and evidence-driven investigation work: two flagship surfaces (an Investigation Terminal and a Tor-integrated Investigation Browser), content-addressed evidence storage, and an append-only cryptographic audit trail.
- Reached V1 Beta: core implementation complete, 348/348 and 310/310 workspace tests passing across two validation phases, a 19-case adversarial security suite, and a candidate ISO verified via automated QEMU cold-boot and live investigation-flow testing. Hardware testing is the current phase.

Mission OS — Privacy-First Linux Distribution

2026 · Open Beta, on hold · Linux, GPLv3

- Designed a privacy-first Linux distribution on Debian Stable and KDE Plasma, hardened automatically at install with no manual post-install script needed.
- Shipped an Open Beta in August 2026 using a customized Calamares installer with Rust-based system services applied automatically after install. Development is paused while other products take priority.

ADDITIONAL PROJECTS

Project Ascent — 25-level 2D precision platformer, Godot 4 · shipped to itch.io with an 18-suite automated test process

Obscura — Luau AST toolkit for Roblox developers · MIT · v1.0.0, 340/340 tests passing

TECHNICAL SKILLS

LANGUAGES	C#, Rust, TypeScript, C / C++, JavaScript, Lua / Luau
FRAMEWORKS	.NET 8, WPF, Node.js, Astro, Qt / QML, Debian Linux
SECURITY	AES-256-GCM, Argon2id, LUKS2, TPM, Secure Boot, Windows Hello
TOOLS	Git / GitHub Actions, VS Code, FFmpeg, Cloudflare Pages

EDUCATION

B.Sc. Animation — D. Y. Patil University, Nerul, Navi Mumbai · 2025 – 2028, second year in progress